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6.046J

Design and Analysis of Algorithms 6.046J — Design and Analysis of Algorithms May

2026 **EXTREMELY HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (2 points) Prove the CAP theorem: a distributed data store cannot simultaneously provide more than two of Consistency, Availability, and Partition tolerance. Formalize the theorem using a proof by contradiction with atomic registers.
2. (6 points) Prove the time hierarchy theorem: for time-constructible functions $f(n)$, there exists a language decidable in $O(f(n))$ time but not in $o(f(n)/\log f(n))$ time.
3. (3 points) Prove the Cook-Levin theorem: SAT is NP-complete. Construct a polynomial-time reduction from any NP language L to SAT.
4. (7 points) Prove the Adleman theorem: $\text{BPP} \subseteq \text{P}/\text{poly}$. Show that any randomized polynomial-time algorithm can be derandomized with polynomial-size advice.
5. (3 points) Prove the Karger-Stein randomized min-cut algorithm runs in $O(n^2 \log^3 n)$ time and succeeds with high probability. Analyze the probability that a particular min-cut survives the contraction process.
6. (3 points) Prove that PP (Probabilistic Polynomial time) is closed under complement and relate it to PH via Toda's theorem. Show that $\text{PH} \subseteq \text{P}^{\#\text{P}}$.
7. (2 points) Prove the Chernoff bound: for independent Bernoulli random variables $X_1, \dots, X_n$ with $\Pr[X_i = 1] = p_i$, let $X = \sum X_i$ and $\mu = \mathbb{E}[X]$. Show that $\Pr[X \geq (1 + \delta)\mu] \leq (\frac{e^\delta}{(1+\delta)^{1+\delta}})^\mu$ for all $\delta > 0$.
8. (4 points) Prove that $\text{RL} \subseteq \text{SC}$ (randomized logspace is in Steve's class) using the Nisan generator and showing that RL can be derandomized with polynomial seed length.
9. (3 points) Prove that directed Steiner tree is NP-hard by reducing from set cover. Then show it is approximable within $O(\log^2 n)$.
10. (5 points) Prove the space hierarchy theorem: for space-constructible $f(n) = \Omega(\log n)$, $\text{SPACE}[f(n)] \subsetneq \text{SPACE}[f(n) \log f(n)]$.

11. (5 points) Prove that EXP^{NP} contains languages that require exponential-size circuits (Karp-Lipton theorem variant). Show that if $\text{NP} \subseteq \text{P/poly}$ then $\text{PH} = \Sigma_2^P$.
12. (6 points) Prove that the minimum distance of a linear code can be computed using the parity check matrix and show that the decisional version is NP-complete (Berlekamp-Welch theorem variant).
13. (3 points) Prove type soundness for the simply typed lambda calculus using the progress and preservation lemmas. Extend the proof to System F with parametric polymorphism.
14. (2 points) Prove that the Unique Games Conjecture implies the hardness of approximating Max-Cut beyond the Goemans-Williamson constant $\alpha_{GW} \approx 0.878$.
15. (4 points) Prove the PCP theorem: $\text{NP} = \text{PCP}(\log n, 1)$. Construct a verifier for SAT that reads $O(1)$ bits of the proof and uses $O(\log n)$ random bits.
16. (7 points) Prove that the amortized cost of a sequence of n operations in a Fibonacci heap is $O(\log n)$ for extract-min and $O(1)$ for all other operations, using the potential function method.
17. (3 points) Prove correctness of the Cheney garbage collection algorithm using a formal argument about the copy phase preserving the reachability graph.
18. (4 points) Prove the Goldreich-Levin theorem: if f is a one-way function, then there exists a one-way function g and a hardcore predicate b such that $b(x)$ is hard to predict given $g(x)$.
19. (9 points) Prove that one-way functions exist if and only if $\text{P} \neq \text{NP}$. Construct a one-way function based on the hardness of SAT, or show that the existence of OWFs is equivalent to pseudorandom generators.
20. (3 points) Prove the parallel repetition theorem for interactive proofs: if a verifier accepts with probability p in a single round, then the verifier accepts in n parallel repetitions with probability at most $p^{\Omega(n)}$ (for certain protocols).
21. (2 points) Prove the VC-dimension bound for PAC learning: if a hypothesis class $\mathcal{H}$ has VC-dimension d , then the sample complexity for (ϵ, δ) -PAC learning is $m \geq \Omega(\frac{1}{\epsilon}(d + \log(1/\delta)))$. Provide the upper bound as well.
22. (6 points) Prove that if $\text{P} = \text{NP}$, then $\text{BPP} = \text{P}$. Conversely, show that $\text{BPP} \subseteq \text{EXP}$ and discuss the relationship between P and BPP .
23. (3 points) Prove that the matrix multiplication exponent ω satisfies $2 \leq \omega < 2.373$ by analyzing the Coppersmith-Winograd algorithm and its subsequent improvements.

24. (3 points) Prove that 3-SAT reduces to Hamiltonian Cycle in polynomial time. Construct the explicit gadget for each clause and variable, showing that a satisfying assignment corresponds to a Hamiltonian cycle in the constructed graph.
25. (2 points) Prove that graph isomorphism is in $\mathbf{NP} \cap \mathbf{coAM}$. Show that **GI** is unlikely to be NP-complete (if it were, **PH** collapses).

Solutions

Solution 1:

Assume all three hold. During a partition, two nodes cannot communicate. A write to node A must be visible on a read from node B (consistency). But node B must respond (availability) without contacting A (partition). So B has either stale data (no consistency) or must wait (no availability), contradiction. This shows at most two properties can hold simultaneously.

Solution 2:

Construct a diagonalizing TM D that on input x runs M_x on x for $f(|x|)$ steps and outputs the opposite answer. If D runs in $o(f(n)/\log f(n))$ time, we get a contradiction for all sufficiently large x . The $\log f(n)$ overhead comes from simulation costs: D must simulate M_x which has a different tape alphabet and number of tapes.

Solution 3:

Let $L \in \text{NP}$ via NTM M running in time $p(n)$. Construct a boolean formula ϕ with variables encoding the computation tableau of M on input x of size n . ϕ has clauses for: (1) initial configuration, (2) valid transitions using $O(p(n)^3)$ variables, (3) accepting state. ϕ is satisfiable iff M accepts x . The reduction runs in polynomial time, proving SAT is NP-hard.

Solution 4:

Let $L \in \text{BPP}$ with TM M using $m(n)$ random bits. For each input size n , the set of bad random strings for input x has size $\leq 2^{m(n)}/3$. By union bound over 2^n inputs, the total bad set has size $\leq 2^n \cdot 2^{m(n)}/3 < 2^{m(n)}$ for n large enough. So there exists a good random string r_n that works for all inputs of length n . Provide r_n as advice, giving $\text{BPP} \subseteq \text{P/poly}$.

Solution 5:

Contract edges uniformly until t vertices remain. A fixed min-cut C of size k survives a single contraction step with probability $\geq 1 - 2/t$. By induction, survival probability $\geq \binom{t}{2}/\binom{n}{2}$. Setting $t = n/\sqrt{2}$ and recursing twice gives $O(n^2 \log n)$ expected time. The success probability is $\Omega(1/\log n)$, and repeating $O(\log^2 n)$ times gives high probability.

Solution 6:

PP is closed under complement by definition ($\Pr[M(x) = 1] > 1/2$ vs $\Pr[M(x) = 1] \leq 1/2$). Toda's theorem: $\text{PH} \subseteq \text{P}^{\#\text{P}}$ by showing $\Sigma_k^{\text{P}} \subseteq \text{BPP}^{\oplus \text{P}}$ (using random hashing) and $\text{BPP}^{\oplus \text{P}} \subseteq \text{P}^{\#\text{P}}$ (since counting parity of solutions is in $\#\text{P}$). This shows counting is harder than the entire polynomial hierarchy.

Solution 7:

Use the moment generating function: $\Pr[X \geq t] \leq \min_{\lambda > 0} e^{-\lambda t} \mathbb{E}[e^{\lambda X}] = \min_{\lambda > 0} e^{-\lambda t} \prod \mathbb{E}[e^{\lambda X_i}]$. Since $\mathbb{E}[e^{\lambda X_i}] = 1 + p_i(e^\lambda - 1) \leq \exp(p_i(e^\lambda - 1))$, we get $\Pr[X \geq (1 + \delta)\mu] \leq (\frac{e^\delta}{(1+\delta)^{1+\delta}})^\mu$ by optimizing $\lambda = \ln(1 + \delta)$.

Solution 8:

Nisan's pseudorandom generator for logspace: $G : \{0, 1\}^{O(\log^2 n)} \rightarrow \{0, 1\}^n$ fools

any logspace machine. The construction uses the INW generator (impagliazzo-Nisan-Wigderson) based on extractors. By enumerating all $2^{O(\log^2 n)}$ seeds, we can simulate RL in SC (polynomial time, polylog space). This gives $RL \subseteq SC \subseteq P$.

Solution 9:

Given set cover instance $(U, \mathcal{S}, k)$, construct a directed graph with source s , set vertices s_j , element vertices u_i , edges $s \rightarrow s_j$ (cost 1), $s_j \rightarrow u_i$ if $u_i \in S_j$ (cost 0). Steiner tree connecting s to all u_i uses $\leq k$ set vertices iff cover of size k exists. The approximation uses randomized rounding and the junction tree theorem.

Solution 10:

Diagonalization: construct a language L decidable in $O(f(n) \log f(n))$ space but not in $o(f(n))$ space. The diagonalizing machine D on input x simulates M_x on x using at most $f(|x|)$ space and if M_x halts within $2^{O(f)}$ steps, outputs the opposite. The simulation uses a counter of $O(\log f)$ bits. The $\log f(n)$ factor is needed for configuration counting to detect infinite loops.

Solution 11:

If $NP \subseteq P/\text{poly}$, then SAT has polynomial-size circuits. By the Karp-Lipton theorem, $\Pi_2^P \subseteq \Sigma_2^P$, collapsing PH to Σ_2^P . The proof: for any Π_2^P language, the existential player guesses the circuit for SAT and uses it to answer the universal quantifier. This shows $PH \subseteq \Sigma_2^P$, hence $PH = \Sigma_2^P$.

Solution 12:

For a linear $[n, k]$ code with parity check matrix H , the minimum distance $d = \min_{c \neq 0, Hc^T=0} w(c)$. Decisional problem (given G and w , is $d \leq w$?) is NP-complete via reduction from 3SAT, by constructing a code whose parity checks encode clause satisfaction. For a linear code over $\mathbb{F}_q$, finding minimum weight codeword is NP-hard.

Solution 13:

Progress: if $\vdash e : \tau$ and e is closed, then e is a value or $e \rightarrow e'$. Proof by induction on typing derivation. Preservation: if $\vdash e : \tau$ and $e \rightarrow e'$, then $\vdash e' : \tau$. Proof by induction on the reduction relation with substitution lemma. For System F, add type abstraction and application cases, with type substitution preserving the subject reduction property.

Solution 14:

Assuming UGC, Khot et al. showed that Max-Cut is NP-hard to approximate within any factor greater than α_{GW} . The reduction: start from a unique game instance, construct a graph using long-code encoding of labels. A labeling that satisfies $1 - \varepsilon$ of unique constraints gives a cut with value $\alpha_{GW} + o(1)$. Dictatorship testing and Fourier analysis are used to prove completeness and soundness.

Solution 15:

$(\supseteq)$ is trivial. $(\subseteq)$: Build a PCP verifier for SAT. Start from the NP-complete constraint satisfaction problem with constant degree (via expander graphs). Apply the gap amplification step (repeatedly replace constraints by their AND using composition). The final verifier reads $O(1)$ proof bits. The theorem shows that approximate solutions to NP problems remain hard.

Solution 16:

Define $\Phi = \text{number of trees} + 2 \cdot \text{number of marked nodes}$. Insert: add one tree, $\Delta\Phi = 1$, actual cost $O(1)$, amortized $O(1)$. Decrease-key: cut and cascade, each cut adds a tree but may clear marks. Extract-min: $O(\text{trees} + \text{degree})$ actual cost. By the rank bound $\text{degree} = O(\log n)$, and trees decrease by at most $O(\log n)$, giving amortized $O(\log n)$.

Solution 17:

Cheney's algorithm uses a semi-space collector. Maintain *fromspace* and *tospace*. The scan pointer iterates through *tospace* while the alloc pointer advances. Invariant: all reachable objects from roots have been copied or are in the queue. Each edge traversed preserves connectivity: $\forall x$ reachable from root, x is either in *tospace* (scanned or unscanned) or reachable from unscanned objects. Termination: when $\text{scan} = \text{free}$, all reachable objects are in *tospace*.

Solution 18:

Define $g(x, r) = (f(x), r)$ for $|x| = |r|$. The hardcore predicate $b(x, r) = x \cdot r \bmod 2$. If an adversary predicts b with advantage ε , construct an inverter for f : use a clever reconstruction (Goldreich-Levin list decoding) that, given $f(x)$ and oracle access to the predictor, outputs a list containing x with high probability. The reconstruction uses pairwise independent sampling and majority decoding.

Solution 19:

$(\Rightarrow)$: If OWF f exists, then inverting f is in NP (guess a preimage and verify) but not in P by definition. So $P \neq NP$. $(\Leftarrow)$: Unconditionally constructing OWF from $P \neq NP$ is open. However, using Levin's construction, we can define $f(M, x, 1^t) = (M, x, 1^t)$ where M outputs the result of t steps, which is one-way if $P \neq NP$ under certain assumptions.

Solution 20:

For XOR games, Raz's parallel repetition theorem shows the value decays exponentially: $\omega(G^{\otimes n}) \leq \omega(G)^{cn}$ where c depends on the game. The proof uses information theory: the winning probability for each coordinate is correlated only through the initial shared randomness, and the protocol cannot coordinate between coordinates without increasing communication.

Solution 21:

Lower bound: construct d shattered points. The adversary picks a target concept uniformly from 2^d dichotomies. To learn, we must see each point with label to distinguish. By the Sauer-Shelah lemma, $|\mathcal{H}(m)| \leq (em/d)^d$. Setting sample size $m = O(\frac{1}{\varepsilon}(d \log(1/\varepsilon) + \log(1/\delta)))$ gives the upper bound via the ERM principle.

Solution 22:

If $P = NP$, then $PH = P$ (polynomial hierarchy collapses). Since $BPP \subseteq \Sigma_2^P \cap \Pi_2^P$ (by Sipser-Gacs theorem), we get $BPP \subseteq P$. The inclusion $BPP \subseteq EXP$ follows from exhaustive simulation of the probabilistic computation over all random bits. Whether $P = BPP$ is open but believed true due to pseudorandom generators.

Solution 23:

Strassen's $O(n^{2.807})$ algorithm uses the identity $M_2(2) = 7$ giving $\omega = \log_2 7$. Coppersmith-Winograd uses the laser method and bilinear algebra to achieve $\omega < 2.376$. The key idea: construct a family of tensors T_n with border rank r_n and combining them recursively. The asymptotic inequality $\omega < \log_{n^{2/3}}(r_n)$ yields improved bounds. The current best $\omega < 2.37286$ uses refined laser method analysis.

Solution 24:

For each variable x_i , create a chain of vertices $v_{i,1}, \dots, v_{i,k}$ with edges forming a path that can be traversed left-to-right (true) or right-to-left (false). For each clause $C_j = (\ell_1 \vee \ell_2 \vee \ell_3)$, insert a clause vertex connected to the corresponding literal vertices in the variable chains. A Hamiltonian cycle exists iff we can pick a consistent direction for each variable chain that hits all clause vertices. This is the standard NP-completeness reduction.

Solution 25:

$\text{GI} \in \text{NP}$: a permutation is a polynomial-time verifiable certificate. $\text{GI} \in \text{coAM}$: the verifier sends a random permutation of one graph; the prover must identify which graph. The interactive protocol has soundness $1/2$. If GI were NP-complete, then $\text{PH} = \Sigma_2^P$ (Boole'an-Karp'n theorem) because $\text{coNP} \subseteq \text{AM}$ and AM collapses if $\text{coNP} \subseteq \text{AM}$.